MILTON CABALLERO

PH.D. CANDIDATE, DATA SCIENCE & ANALYTICS | OPTIMIZATION UNDER UNCERTAINTY

The University of Oklahoma, Norman, OK
miltons1525@gmail.com | +1 (405) 565-0665 | linkedin.com/in/miltoncab | miltoncaballero.github.io

RESEARCH INTERESTS

Optimization and data-driven modeling for resilient, sustainable interdependent networks. My work centers on synchromodal network science under demand and disaster-induced uncertainty, distributionally robust frameworks for multi-hazard risk mitigation, and graph-theoretic formulations that improve the performance and adaptability of critical infrastructure systems.

EDUCATION

The University of Oklahoma — Norman, OK, USA
Ph.D. in Data Science and Analytics — GPA 4.00/4.00 — Jun 2024 – Present (Expected Graduation: Dec 2027)

- Advisor: Dr. Andrés D. González, Netsys Research Lab
- Coursework distinction (A) in Statistics, Data Analysis, and Optimization

The University of Oklahoma — Norman, OK, USA
M.S. in Data Science and Analytics — GPA 4.00/4.00 — Jun 2024 – Dec 2025

Industrial University of Santander (UIS) — Bucaramanga, Colombia
B.S., Double Major: Industrial Engineering and Mathematics — GPA 3.74/4.00 — Sep 2016 – Dec 2022

- Recognized as a distinguished student for sustained academic performance across both majors
- Research seminars with the Finance and Management group and the Optimization and Organization of Production, Administrative and Logistics Systems (OPALO) group

The University of Oklahoma — Norman, OK, USA
Undergraduate Exchange Program, Industrial Engineering and Finance — GPA 4.00/4.00 — Jan 2022 – Jul 2022

PUBLICATIONS

Published

1. Caballero, M., González, A.D. (2026). "Synchromodal Bi-Objective Two-Stage Stochastic Optimization for Container and Power Unit Allocation." *Transportation Research Part E* — Volume 216, Article 105085.

In Preparation

2. Caballero, M., González, A.D., Borrero, J. (2026). "Data-Driven Distributionally Robust Minimum-Cost Flow Problem under Ripple Effect Disruptions." *Target: European Journal of Operational Research.*
3. Caballero, M., González, A.D. (2026). "Uncertainty Propagation Affecting Decisions in Network Systems: Optimizing Network Resilience with Targeted Arc Additions."
4. Caballero, M., González, A.D. (2026). "Scarcity Regimes and the Timing of Fairness in Disrupted Infrastructure Networks." *Target: Omega SI.*
5. Caballero, M., González, A.D. (2026). "Street-Turn-Enabled Synchromodal Planning: The Environmental Value of Coordination at the Drayage–Trucking Interface." *Target: Sustainability Analytics and Modeling SI.*

CONFERENCE PRESENTATIONS

1. Caballero, M., González, A.D. "Vehicle Routing Problems in Pickup and Delivery Services" *Institute of Industrial and System Engineers (IISE) Annual Conference & Expo*, Arlington, Texas, 2026.

2. Caballero, M., González, A.D. "Optimization under Uncertainty in Transportation, Logistics and Supply Chains; Freight Transportation Session" *24th Conference of the International Federation of Operational Research Societies (IFORS)*, Vienna, Austria, 2026.
3. Caballero, M., González, A.D., Borrero, J.S. "Resilient Infrastructure and Communities" *2026 Institute for Operations Research and the Management Sciences (INFORMS) Annual Meeting*, San Francisco, California, 2026.

RESEARCH AND TEACHING EXPERIENCE

Netsys Research Lab, The University of Oklahoma — Norman, OK, USA
Graduate Research Assistant — Jun 2024 – Present

- Lead author on four manuscripts developing stochastic and distributionally robust optimization models for network systems under demand and disruption uncertainty
- Build and validate large-scale mathematical programs in Python and Gurobi, including bi-objective two-stage stochastic formulations and data-driven ambiguity sets
- Additional focus areas: stochastic methods for uncertainty management, supply chain networks, graph theory and fairness

School of Industrial & Systems Engineering, The University of Oklahoma — Norman, OK, USA
Graduate Teaching Assistant — Jun 2025 – Present
Courses: Systems Optimization; Advanced Systems Optimization; Quality and Reliability Engineering

- Deliver lectures and lead discussion sections in person and online
- Develop course materials and update their content
- Evaluate student work, provide detailed written feedback, and hold weekly office hours

Accreditation Board for Engineering and Technology (ABET) — Bucaramanga, Colombia
Research Assistant — Sep 2022 – Dec 2023

- Conducted a diagnostic analysis of the Industrial Engineering program accreditation process and identified the data required to support it
- Designed and implemented a Microsoft Power BI tool for managing and visualizing accreditation data, built to the requirements of the ABET team

PRIOR PROFESSIONAL EXPERIENCE

Amazon — Bucaramanga, Colombia
Shipping and Delivery Support (SDS) Agent — Sep 2020 – Sep 2021

- Resolved logistics exceptions across the delivery network under high-volume service targets
- Managed shipment tracking records in business databases

Incodim S.A.S. — Bucaramanga, Colombia
Administrative Assistant — Feb 2017 – Aug 2019

- Managed resource and personnel accounting for civil works projects serving local communities
- Prepared public bids and project plans for incoming contracts

FELLOWSHIPS AND AWARDS

- **Robberson Travel Grant** — The Graduate College, University of Oklahoma. Awarded to present at the 24th Conference of the International Federation of Operational Research Societies (IFORS), Vienna, Austria
- **Gallogly College of Engineering Graduate Fellowship** — University of Oklahoma. Awarded for an outstanding academic record within the college

- **Gallogly College of Engineering Graduate Student Travel Award** — Awarded to present at the IISE Conference and Expo 2026, Arlington, Texas
- **Distinguished Student** — Industrial University of Santander, for academic performance while completing a double major in Industrial Engineering and Mathematics

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, pandas, NetworkX, scikit-learn, Keras, Matplotlib, Seaborn), SQL, R, C/C++
Optimization	Gurobi, AMPL, CPLEX, GAMS
Statistical & Math	MATLAB, SageMath, Minitab, advanced Microsoft Excel
Design & GIS	ArcGIS, AutoCAD (2D/3D), SolidWorks
Tools & Platforms	LaTeX, Power BI, Visio, MS Office, Google Workspace, AWS (incl. CloudWatch)
Languages	English (fluent), Spanish (native), German and French (basic)

PROFESSIONAL MEMBERSHIPS

Institute of Industrial and System Engineers (IISE) 2024 – Present
Student Member

Institute of Operations Research and Management Sciences (INFORMS) 2025 – Present
Student Member

American Society of Engineering Education (ASEE) 2026 – Present
Student Member

Interdependent Networked Community Resilience Modeling Environment (INCORE) 2024 – Present

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

INFORMS OU Student Chapter Norman, OK
Member 2025 – Present

United States Tennis Association (USTA) Oklahoma City, OK
Competitive Tennis Player 2025 – Present

Colombian Student Association (COLSA), University of Oklahoma Norman, OK
Logistics Officer 2024 – Present

OU Cousins International Exchange Program, University of Oklahoma Norman, OK
Student Member 2024 – Present

OU Tennis Club, University of Oklahoma Norman, OK
Student Athlete 2025 – Present

Uplift Running Club, University of Oklahoma Norman, OK
Student Member 2026 – Present

OU Wellness Club, University of Oklahoma Norman, OK
Student Member 2026 – Present

Study Center Association Leadership Team, Industrial University of Santander Bucaramanga, Colombia
Board Member 2022 – 2023